POMPS Presentation: 1.2.55

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Problem Statement

Question:

The minimum number of terms required in e^x series expansion to evaluate at $x = 1$ correct up to 3 decimal places is:

1. 8
2. 7
3. 6
4. 5

Matrix Operator Representation

Let $\mathbf{S}$ be the infinite-dimensional downshift matrix, representing a discrete unit delay operator (z^{-1}).

When multiplied by a vector, it shifts all elements down by one position:

$$\mathbf{S} = \begin{bmatrix} 0 & 0 & 0 & \dots \\ 1 & 0 & 0 & \dots \\ 0 & 1 & 0 & \dots \\ \vdots & \vdots & \ddots & \ddots \end{bmatrix}, \quad \mathbf{S} \begin{bmatrix} a \\ b \\ c \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 \\ a \\ b \\ \vdots \end{bmatrix} \tag{3.1}$$

Ideal System Response

The ideal infinite impulse response of this system is generated by applying the matrix exponential to an impulse vector $\delta = [1, 0, 0, \dots]^T$.

Expanding the Taylor series of $e^{\mathbf{S}}$ directly generates the Maclaurin coefficients:

$$\mathbf{h}_{\text{ideal}} = e^{\mathbf{S}}\delta = \left(\mathbf{I} + \mathbf{S} + \frac{1}{2!}\mathbf{S}^2 + \frac{1}{3!}\mathbf{S}^3 + \dots\right)\begin{bmatrix}1\\0\\0\\\vdots\end{bmatrix} = \begin{bmatrix}1\\1\\1/2!\\1/3!\\\vdots\end{bmatrix} \tag{3.2}$$

Filter Truncation and Error Matrix

Truncating this infinite system to a finite filter length N yields the error matrix, $\mathbf{E}_N$:

$$\mathbf{E}_N = \sum_{k=N}^{\infty} \frac{1}{k!} \mathbf{S}^k \tag{3.3}$$

To measure the maximum size of this error, we apply the induced l_1-norm ($\|\cdot\|_1$), defined as the maximum absolute column sum of a matrix. We use this instead of Frobenius as we have an infinite matrix and this norm looks for worst-case scenario in each column.

Applying the Induced l_1-Norm

Because every column of $\mathbf{S}$ contains exactly one 1 and zeros elsewhere, its column sum is strictly 1, hence $\|\mathbf{S}\|_1 = 1$.

By the multiplicative property of norms ($\|\mathbf{AB}\| \leq \|\mathbf{A}\|\|\mathbf{B}\|$), we establish:

$$\|\mathbf{S}^k\|_1 \leq \|\mathbf{S}\|_1^k = 1^k = 1 \tag{3.4}$$

This allows us to bound the error:

$$\|\mathbf{E}_N\|_1 \leq \sum_{k=N}^{\infty} \frac{\|\mathbf{S}\|_1^k}{k!} = \sum_{k=N}^{\infty} \frac{1}{k!} \tag{3.5}$$

Bounding the Error

Expanding the summation and factoring out the first neglected term, $\frac{1}{N!}$, yields:

$$\|\mathbf{E}_N\|_1 \leq \frac{1}{N!} + \frac{1}{(N+1)!} + \frac{1}{(N+2)!} + \dots \tag{3.6}$$

$$\|\mathbf{E}_N\|_1 \leq \frac{1}{N!}\left(1 + \frac{1}{N+1} + \frac{1}{(N+1)(N+2)} + \dots\right) \tag{3.7}$$

Constructing an Algebraic Bound

To construct a solvable algebraic bound, we replace the increasing denominators with the worst-case constant, N, as $\frac{1}{N+1} < \frac{1}{N}$.

This substitution results in a standard infinite geometric series:

$$\|\mathbf{E}_N\|_1 \leq \frac{1}{N!}\left(1 + \frac{1}{N} + \frac{1}{N^2} + \dots\right) = \frac{1}{N!}\sum_{m=0}^{\infty}\left(\frac{1}{N}\right)^m \tag{3.8}$$

$$\|\mathbf{E}_N\|_1 \leq \frac{1}{N!}\left(\frac{1}{1 - 1/N}\right) = \frac{1}{N!}\left(\frac{N}{N-1}\right) \tag{3.9}$$

Defining Error Tolerance

The problem requires the evaluation to be correct up to 3 decimal places.

To guarantee this precision, the absolute truncation error must be strictly less than half of the place value of the third decimal digit (10^{-3}).

Therefore, our target error tolerance (ϵ) is bounded by:

$$\epsilon \leq \frac{1}{2} \times 10^{-3} = 0.0005 \tag{3.10}$$

Enforcing Tolerance Limits

For 3 decimal places of precision, we apply error tolerance ($\epsilon \leq 0.0005$) on our derived bound:

$$\frac{1}{N!}\left(\frac{N}{N-1}\right) \leq 0.0005 \tag{3.11}$$

Checking values of the filter length N:

$$N = 5: \quad \frac{1}{120}\left(\frac{5}{4}\right) = \frac{1}{96} \approx 0.01042 \quad \text{(Not met)} \tag{3.12}$$

$$N = 6: \quad \frac{1}{720}\left(\frac{6}{5}\right) = \frac{1}{600} \approx 0.00166 \quad \text{(Not met)} \tag{3.13}$$

$$N = 7: \quad \frac{1}{5040}\left(\frac{7}{6}\right) = \frac{7}{30240} \approx 0.00023 \quad \text{(Met)} \tag{3.14}$$

Comparison with Classical Lagrange Bound

The standard method to solve this question is to use Lagrange Error Bound for N terms (highest power $n = N - 1$):

$$R_{N-1}(x) = \frac{e^c}{N!}x^N \quad \text{where } c \in (0, 1) \tag{3.15}$$

Evaluating at $x = 1$ and knowing $e < 3$, the classical upper bound is:

$$R_{N-1}(1) < \frac{3}{N!} \tag{3.16}$$

Testing our derived answer of $N = 7$:

$$R_6(1) < \frac{3}{5040} \approx 0.000595 \tag{3.17}$$

Result: The classical bound fails at our ≤ 0.0005 tolerance. Our derived bound (≈ 0.00023) is significantly tighter, proving $N = 7$ is sufficient. Thus, due to having to guess an upper limit for e, the classical method performs worse than our method. (In fact, taking exact value of e also yields 0.000539, which is still above the threshold)

Conclusion

Thus minimum number of required terms is 7.

The Maclaurin series for e^x evaluated at $x = 1$ is:

$$e^1 = \sum_{k=0}^{\infty} \frac{1}{k!} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots \tag{3.18}$$

If we take N terms, the sum stops at the index $k = N - 1$:

$$S_N = \sum_{k=0}^{N-1} \frac{1}{k!} \tag{3.19}$$

Our target of $N = 7$ terms expands to a 6th-degree polynomial:

$$S_7 = \underbrace{1}_{k=0} + \underbrace{1}_{k=1} + \underbrace{\frac{1}{2!}}_{k=2} + \underbrace{\frac{1}{3!}}_{k=3} + \underbrace{\frac{1}{4!}}_{k=4} + \underbrace{\frac{1}{5!}}_{k=5} + \underbrace{\frac{1}{6!}}_{k=6} = 2.7181 \tag{3.20}$$

Python Implementation I

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import factorial

e_val = np.exp(1)

# start at 3 to avoid div by zero in the N/(N-1) bound
N = np.arange(3, 11)

errs = []
bounds = []

for n in N:
    k = np.arange(n)
    approx = np.sum(1 / factorial(k))

    errs.append(np.abs(e_val - approx))
    bounds.append((1 / factorial(n)) * (n / (n - 1)))

plt.figure(figsize=(9, 5))

plt.plot(N, errs, marker='o', label='Actual Error')
```

Python Implementation II

```
plt.plot(N, bounds, marker='s', ls='--', label='Theoretical Bound')

plt.axhline(y=0.0005, color='r', ls=':', label='Tolerance (0.0005)')
plt.axvline(x=7, color='g', ls=':', alpha=0.5)

plt.annotate('N=7', xy=(7, 0.0005), xytext=(7.3, 0.005),
            arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth
                =5))

plt.yscale('log')
plt.xlabel('Number of terms (N)')
plt.ylabel('Error')
plt.title('Truncation Error for e^x at x=1')
plt.xticks(N)
plt.grid(True, ls="--", alpha=0.4)
plt.legend()

plt.tight_layout()
path='../figs/plot.png'
plt.savefig(path, dpi=300)
print(f"Saved: {path}")
plt.close()
```

Error Plot

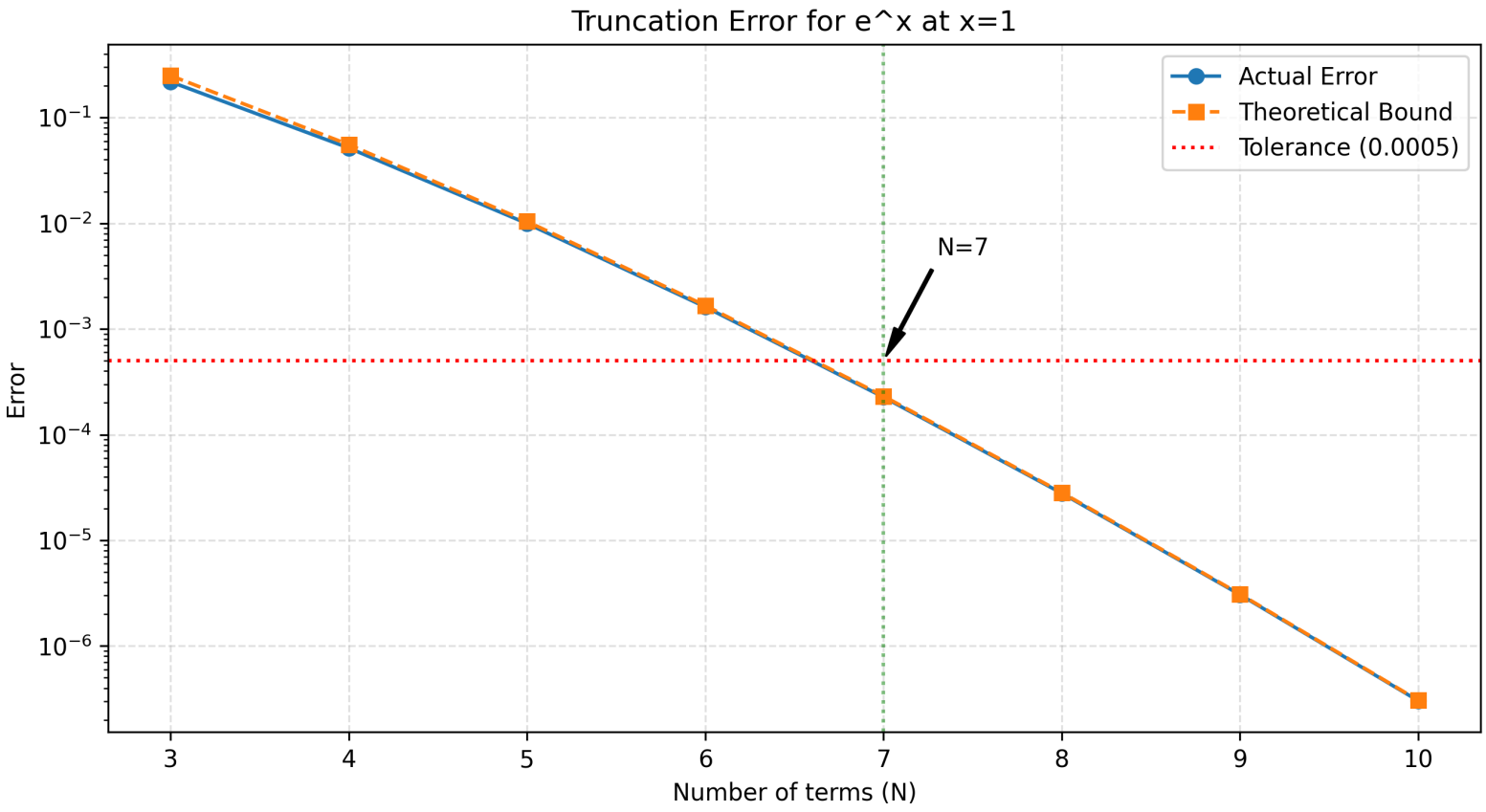

Figure: Error v/s N